Law of Sines and Law of Cosines

Law of Sines

1 In triangle ABC, $A = 40°$, $B = 65°$, and $a = 12$. Find:

a angle C **b** side b (2 d.p.) **c** side c (2 d.p.)

2 Two observers 400 m apart on a straight road both sight a hot-air balloon between them. The angles of elevation are $58°$ and $47°$. Find the distance from the balloon to the first observer (the one with the $58°$ sighting), to the nearest metre.

Law of Cosines

3 In triangle PQR, two sides of length 8 and 11 enclose an angle of $52°$. Find the length of the third side, to two decimal places.

4 A triangle has sides 7, 9, and 12. Find the measure of its largest angle, to the nearest tenth of a degree.

5 Find the area of a triangle with sides 10 and 14 enclosing an angle of $35°$, to two decimal places.

6 A ship sails 30 km due east, then turns and sails 45 km on a bearing of $N\,50°\,E$. Decide which law applies and find the ship's distance from its starting point, to the nearest kilometre.

The ambiguous case (SSA)

7 In triangle ABC, $A = 30°$, $a = 5$, and $b = 9$. Determine how many triangles satisfy these conditions (zero, one, or two), and find all possible measures of angle B.

8 In triangle DEF, $D = 40°$, $d = 10$, and $e = 6$. Determine how many triangles satisfy these conditions, explaining your reasoning.

Choosing the right law

9 For each triangle description, state whether the Law of Sines or the Law of Cosines is the appropriate first tool, and why:

a Two angles and one side are known (AAS).

b Three sides are known (SSS).

c Two sides and the included angle are known (SAS).

10 A triangular plot of land has sides 120 m, 150 m, and 90 m. Find its area, to the nearest square metre, using the fact that you can first find any angle with the Law of Cosines and then use $A = \frac{1}{2}ab\sin C$.

Heron's formula

11 State Heron's formula for the area of a triangle with sides a, b, c and semi-perimeter $s = \frac{a+b+c}{2}$, then use it to find the area of a triangle with sides 13, 14, 15.

12 A triangular garden has sides 9 m, 12 m, and 15 m. Use Heron's formula to find its area, and verify it is a right triangle by checking the Pythagorean relationship.

Navigation and bearing problems

13 A plane flies on a bearing of $N\,60°\,E$ for 200 km, then changes course to a bearing of $S\,80°\,E$ for 150 km. Find the angle between the two legs of the flight, and set up (but do not evaluate) the Law of Cosines expression for the total displacement.

Solving oblique triangles from a diagram

14 In triangle XYZ, $x = 8$, $y = 10$, and angle $Z = 48°$ (between the given sides). Find side z, then find angle X using the Law of Sines, to the nearest tenth of a degree.

Verifying triangle solutions with the angle sum

15 For the triangle solved above ($z \approx 7.55$, $X \approx 51.9°$, $Z = 48°$), find angle Y and verify all three angles sum to $180°$.

Choosing between multiple valid approaches

16 A triangle has $a = 6$, $b = 8$, $c = 10$. Find angle C (opposite the longest side) two different ways: (1) using the Law of Cosines, and (2) recognizing this as a Pythagorean triple.